

Lab 1 Assignment

ECE 525.642 FPGA Design using VHDL

Simple Seven-Segment Display Controller

For this exercise, you will configure the Xilinx field programmable gate array (FPGA) on the Nexy's development board to act as a simple controller for the both of the boards seven-segment displays. The inputs to the top-level design will be a 16-bit `std_logic_vector` input driven by the slider switches and three push buttons to change the state of the seven-segment displays. The output of the top-level design will be an eight bit `std_logic_vector` named `seg7`. This vector will illuminate the seven digit segments and the additional dot digit. Additionally, another 8-bit `std_logic_vector` call the anodes will be used to select which of the seven-segment displays to illuminate. For this implementation, the value of the slider switches SW3-SW0 will be displayed on the seven-segment display using hexadecimal representation (0 through F).

Task 1

Complete `Nexys4DDR_Tutorial.pdf`, as this will guide you through the creation of a project and a simple 7-segment decoder which can be used in the remainder of the lab.

Task 2

Create an entity/architecture pair called `seg7_hex` in a file named `seg7_hex.vhd` (NOTE: It's good practice to name the VHD file the same as the entity name)

This entity should only have two ports: a 4-bit input called `digit` and an 8-bit output called `seg7`. This component will be instantiated in your top-level architecture. You can take inspiration from the Task 1 Tutorial on how to implement this.

Task 3

Create a new top-level entity/architecture pair called `lab1_top` in a file named `lab1_top.vhd`. This new entity will be the main entity for the design and will instantiate the 7-segment decoder that you built in Task 2. This will be the first design built by you that will be synthesized and loaded to the FPGA on the Nexy's board. The entity for `lab1_top` should look like the following code:

```
entity lab1_top is
  Port (
    --Push Buttons
    BTNC : in STD_LOGIC;
    BTND : in STD_LOGIC;
    BTNU : in STD_LOGIC;
    --Switches (16 Switches)
    SW : in STD_LOGIC_VECTOR (15 downto 0);
    --LEDs (16 LEDs)
```



```

    LED : out STD_LOGIC_VECTOR (15 downto 0);
    --Seg7 Display Signals
    SEG7_CATH : out STD_LOGIC_VECTOR (7 downto 0);
    AN : out STD_LOGIC_VECTOR (7 downto 0));
end lab1_top;

```

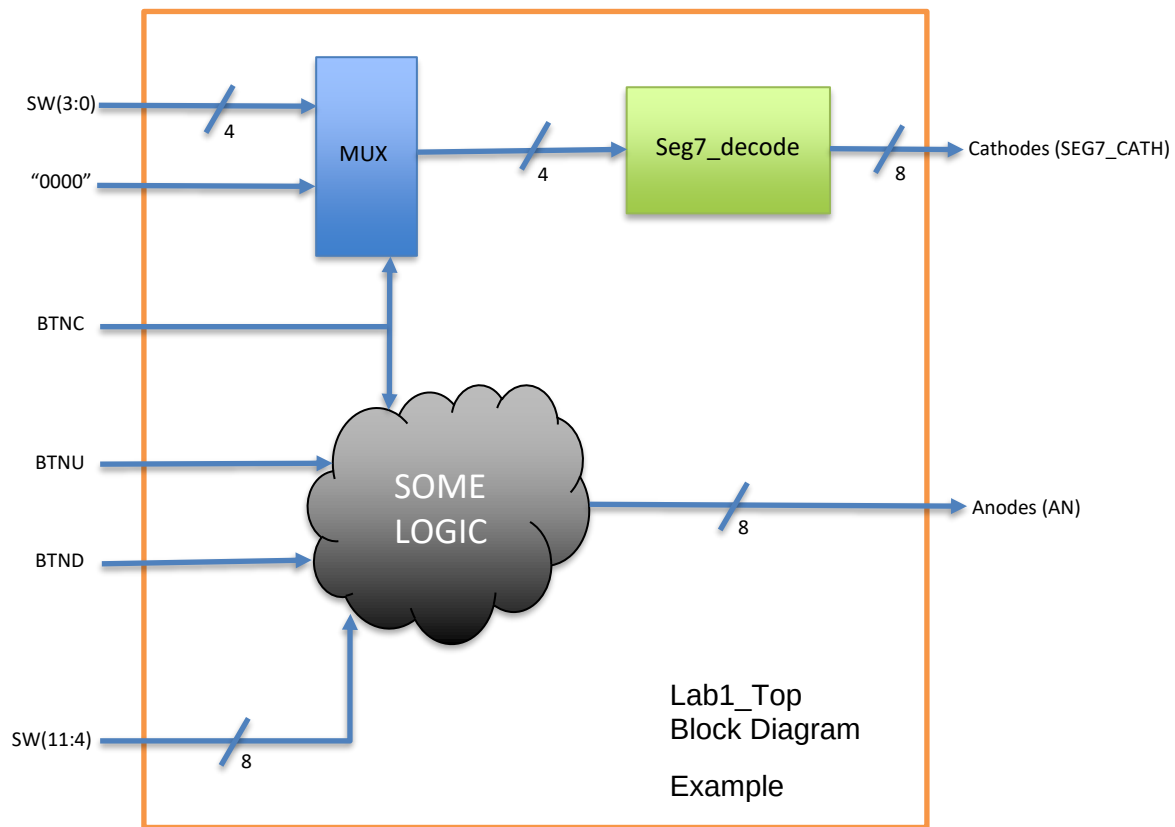
The main component of lab1_top's architecture should be an instantiation of seg7_hex as built in Task 2. You will hook it up to perform the tasks as described below:

1. The input to the seven-segment controller will be the four slider switches SW3-SW0 or all zero's (depending on push buttons below). SW3 is considered to be the most significant bit (msb) of the binary number and SW0 the least (lsb).
2. Normally, with none of the push buttons pressed, the displays will be active depending on the state of SW11 through SW4, which will form an 8-bit binary number. Each bit determines which seven-segment display will be active. SW11 makes the left-most display active, SW10 the one right next to it, and this continues to SW4 which activates the right-most display.
3. When the UP BTN is pressed (**BTNU**), only the upper four displays will be active and showing the SW3 to SW0 value, independent of the values of SW11 through SW4.
4. When the DOWN BTN is pressed (**BTND**), only the bottom four displays will be active and showing the SW3 to SW0 value, independent of the values of SW11 through SW4.
5. When the CENTER BTN is pressed (**BTNC**), all of the displays will be active and showing all zeros, independent of the values of SW11 through SW4 and independent of the digit selection of SW3 to SW0.
6. The green LEDs, labeled LD15 through LD0 will be lit up on the board based on which switch SW15 through SW0 is set to the HIGH position. This should occur ALL the time.

NOTE: Assume only one push button is pressed at any given time.

A small block diagram of how this might be implemented is illustrated below. Your design should consist of two entity/architecture pairs: seg7_hex and lab1_top, no others. In other words, a block on the diagram below doesn't necessarily mean a new VHDL entity, just some logic. The top_level entity/architecture pair (lab1_top) should instantiate the seven-segment encoder.





Task 4

Set the pin assignments for the ports of lab1_top to the appropriate Xilinx FPGA pins and save this file as a XDC. I would recommend starting with the Nexy's Master XDC File and just commenting out all the unused signals. Keeping the naming consistent with the Master XDC file will make it easier to share designs throughout this course. The only change I make in my top level example above is naming the SEG7_CATH as an array instead of independent signals. I recommend making this change as well. After that, synthesize, generate a programming file, and upload this file to the board using the Hardware Manager. Test your design to verify that it works. Remember, not having it work on the first try is normal and this just means take a step back, look at your logic, revise, and repeat the same build steps.

To Submit

Submit the following files zipped up together in a file called Lab1_<last_name>.zip:

- All VHD files for the Lab
- The XDC constraints file
- BIT file for programming the FPGA
- Synthesis Results as Word or Text file

*No report is submitted for Lab 1 unlike future labs

Note: The Grading sheet provided on the next page provides a good overview of the key points of the design and what the instructor will look for when grading. Please note that this is a guide and isn't comprehensive. Please read over the lab assignment carefully to verify all the proper documents are submitted.



Grading Sheet

Lab 1/Module 1: Simple 7-segment display

Student:

Grade	Out of	Notes
	100	
	25	Seven segment encoder module correctly implemented
	5	Value of SW3-SW0 shows up on seven-segment display
	10	Values displayed on seven segment display should only be hexadecimal
	10	Green LEDs light up according to which slider switch is in HIGH position
	25	Correct Operation of Push Buttons
		Slider switches SW3-SW0 represent input digit
		1. With no buttons pressed, the digit is displayed in any display that has the corresponding slider switch set HIGH from SW11 to SW4.
		2. When the UP BTN is pressed (BTNU), only the upper four displays will be active and showing the SW3 to SW0 value
		3. When the DOWN BTN is pressed (BTND), only the bottom four displays will be active and showing the SW3 to SW0 value
		4. When the CENTER BTN is pressed (BTNC), all of the displays will be active and showing all zeros
	20	Well Documented and appropriate VHDL Code
	5	Synthesis report with device resource utilization summary

